

Rheinhafen Combined Cycle Gas Turbine (CCGT) - Plant Operating Manual

Rheinhafen Combined Cycle Gas Turbine (CCGT) - Plant Operating Manual

Siemens SGT5-8000H Gas Turbine + SST5-5000 Steam Turbine

Nameplate Capacity: 430 MW (net) | Commissioned: 2019

1. Plant Configuration

1.1 Gas Turbine Island

Parameter	Value
Manufacturer	Siemens Energy
Model	SGT5-8000H (H-class)
Fuel	Natural gas (pipeline quality, Wobbe Index 47-53 MJ/m ³)
Gross output (GT only)	295 MW at ISO conditions
Turbine inlet temperature	1,500 C (design)
Pressure ratio	21:1
Exhaust temperature	625 C
Exhaust mass flow	820 kg/s

1.2 Heat Recovery Steam Generator (HRSG)

Parameter	Value
Type	Triple-pressure with reheat
HP steam	170 bar / 565 C
IP steam	35 bar / 565 C (reheat)
LP steam	4.5 bar / 250 C

1.3 Overall Plant Performance

Parameter	ISO Conditions	Summer (35C)	Winter (-5C)
Net output	430 MW	395 MW	455 MW
Net efficiency (LHV)	58.0%	56.2%	59.1%
Heat rate (Btu/kWh)	5,880	6,069	5,770
CO2 intensity	0.349 tCO2/MWh	0.360 tCO2/MWh	0.342 tCO2/MWh

2. Operating Modes & Constraints

2.1 Load Range

Mode	Load Range	Notes
Full load	380-430 MW	Normal operating range
Part load	170-380 MW	Efficiency penalty applies
Minimum stable load	170 MW (40%)	Below this: flame instability

2.2 Ramp Rates

Condition	Ramp Up	Ramp Down
Normal operation	20 MW/min	20 MW/min
Fast ramp (GT only)	30 MW/min	30 MW/min
Frequency response	+/- 5% in 30s	+/- 5% in 30s

2.3 Start-Up Times & Costs

Start Type	Condition	Time to Min Load	Time to Full Load	Total Start Cost
Hot start	<8h offline	30 min	60 min	45,000 EUR

Start Type	Condition	Time to Min Load	Time to Full Load	Total Start Cost
Warm start	8-48h offline	90 min	150 min	85,000 EUR
Cold start	>48h offline	240 min	360 min	145,000 EUR

2.4 Minimum Run/Down Times

Parameter	Value	Reason
Minimum run time	4 hours	Thermal stress limits on HRSG headers
Minimum down time	4 hours	Cooling requirements before restart
Maximum starts per day	2	Lifetime management

3. Efficiency & Part-Load Performance

3.1 Willans Line

$\eta(P) = \eta_{\max} * (1 - k * (1 - P/P_{\max})^2)$
 $\eta_{\max} = 0.580$, $k = 0.15$, $P_{\max} = 430$ MW

3.2 Part-Load Efficiency Table

Load (MW)	Efficiency (%)	Heat Rate (Btu/kWh)	SRMC (EUR/MWh)
430	58.0	5,880	68.4
400	57.7	5,912	68.8
350	57.0	5,986	69.7
300	55.9	6,103	71.1
250	54.4	6,272	73.1
200	52.4	6,511	76.0
170	50.8	6,717	78.4

(SRMC calculated at TTF=32 EUR/MWh, ETS=65 EUR/tCO₂)

3.3 Ambient Temperature Correction

- Output: -0.7% per degree C above 15C ISO
- Efficiency: -0.5% per degree C above 15C ISO

3.4 Short-Run Marginal Cost Formula

$$\text{SRMC} = \text{Gas_Price} / \text{Efficiency} + \text{CO2_Price} * (0.202 / \text{Efficiency}) + \text{VOM}$$
$$\text{VOM} = 2.70 \text{ EUR/MWh}$$

4. Fuel System

Parameter	Value
Gas source	TransnetBW high-pressure grid (70 bar)
Gas quality	H-gas (Wobbe 47-53 MJ/m ³ , CH ₄ > 85%)
Maximum consumption	74,000 Nm ³ /h at full load
Gas calorific value	39.0 MJ/Nm ³ (HHV) / 35.2 MJ/Nm ³ (LHV)
Conversion	1 MWh _{th} = 3.412 MMBtu = 3.6 GJ

5. Emissions & EU ETS

Parameter	Value
Emission factor (gas)	0.202 tCO ₂ /MWh _{th}
Specific emissions (full load)	0.349 tCO ₂ /MWh _e
Specific emissions (min load)	0.398 tCO ₂ /MWh _e
EU ETS Phase IV	No free allocation for power generation
Carbon cost at full load	0.349 * ETS_price EUR/MWh

6. Trading-Relevant Operating Procedures

6.1 Pre-Start Checklist

1. Confirm plant status (hot/warm/cold) with control room
2. Calculate start cost + minimum run time revenue requirement
3. Verify gas nomination is possible (check gas day timing)
4. Check ambient temperature for output correction
5. Break-even price: $(\text{Start_Cost} / (\text{Min_Run_Hours} * \text{Capacity})) + \text{SRMC}$

6.2 Dispatch Optimization Rules

- Never start for less than minimum run time (4 hours)
- Never operate below minimum stable load (170 MW)
- Ramp constraint: max 20 MW per minute
- If selling 430 MW for hour H, must be at 430 MW by start of H

- Ramp from 170 to 430 MW takes $(430-170)/20 = 13$ minutes

6.3 Gas Nomination Timing

- Day-ahead nomination: by 14:00 CET D-1
- Within-day nomination: 2-hour lead time
- Renomination: every 2 hours during gas day
- Imbalance tolerance: +/- 5% of nominated volume